

Xiurui Zhao (赵修瑞)

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Fellowships & Appointments

California Institute of Technology, Pasadena, U.S.	2025-
Postdoctoral Scholar Research Associate, Advisor: Prof. Fiona Harrison, Dr. Daniel Stern	
University of Illinois Urbana-Champaign, Urbana, U.S.	2023-2025
Postdoctoral Research Associate, Advisor: Prof. Yue Shen	
Center for Astrophysics Harvard & Smithsonian, Cambridge, U.S.	2021-2023
Postdoctoral Fellow, Advisors: Dr. Francesca Civano, Dr. Martin Elvis	
Center for Astrophysics Harvard & Smithsonian, Cambridge, U.S.	2020-2021
Pre-Doctoral Fellow, Advisor: Dr. Francesca Civano	

Education

Clemson University, Clemson, U.S.	2016-2021
Ph.D. in Astrophysics, Advisor: Prof. Marco Ajello	
Dissertation: <i>Heavily Obscured Active Galactic Nuclei in NuSTAR Era</i>	
Lanzhou University, Lanzhou, China	2012-2016
B.Sc. in Physics, Cuiying Honors College, China's Top-Notch Undergraduate Training Program	

Honors and Awards

Clemson University Level Outstanding Graduate Researcher (2 winners each year)	2021
Science College Outstanding Graduate in Discovery	2021
Physics Department Graduate Research Assistant	2021
SAO Predoctoral Fellowship	2020-2021

Accepted Scientific Proposals as PI

29 accepted X-ray/optical/sub-mm proposals with **\$660k** grant as PI.

- **X-ray (2600 ks)**

- **NuSTAR Cycle 12 (Large, 500 ks NuSTAR + 125 ks XMM, \$130k)** **2026**
"Systematically Probing the Physics of AGN Coronae Using Luminous, High-redshift Quasars"
- **NuSTAR Cycle 12 (ToO, 140 ks NuSTAR + 28 ks Swift, \$85k)** **2026**
"A Dynamic View of AGN Coronae with NuSTAR"

- **Chandra DDT (40 ks)** **2025**
“X-ray Monitoring of the JWST-NEXUS Field”
- **XMM-Newton DDT (25 ks)** **2025**
“First X-ray Survey on the JWST-NEXUS Field”
- **NuSTAR Cycle 11 (ToO, 100 ks NuSTAR + 28 ks Swift, \$85k)** **2025**
“Probing the electron populations in AGN Corona with NuSTAR”
- **Swift-XRT ToO (18 ks)** **2025**
“Measure the current flux of eROSITA detected high-z quasars for NuSTAR follow-up”
- **NuSTAR Cycle 9 (Large, 500 ks NuSTAR + 142 ks XMM, \$130k)** **2023**
“Systematically Constraining the AGN Coronal Properties with NuSTAR Using a Sample of Luminous, High-redshift Quasars”
- **NuSTAR Cycle 8 (Large, 600 ks NuSTAR + 195 ks XMM, \$150k)** **2022**
“Constraining the Properties of AGN Coronae using a Sample of Luminous, High-redshift Quasars with NuSTAR”
- **NuSTAR Cycle 7 (100 ks NuSTAR + 60 ks XMM, \$90k)** **2021**
“Unveiling with NuSTAR the most powerful, heavily obscured, quasar ever discovered in X-rays”
- **Swift-XRT Cycle 19 (18 ks)** **2022**
“Build a Sample of Luminous, High-redshift Quasars to Constrain the Properties of AGN Coronae”
- **Swift-XRT ToO (3 ks)** **2021**
“Measure the X-ray flux of a rare coronal line event quasar exhibiting another optical flare”
- **Optical (15 nights)**
- **Keck OSIRIS-LGS (0.25 night)** **2026B**
“Roman Precursor: Studying Metallicity Dependence of Microlensing Planet-host M Dwarf”
- **Palomar-P200 NGPS (3+1 nights)** **2026A/B**
“A Complete Census of Swift-BAT Hard X-ray Survey Detected AGN”
- **SOAR 4m Goodman (1.5 night)** **2025B**
“A Complete Census of Swift-BAT Hard X-ray Survey Detected AGN”
- **SOAR 4m Goodman (0.6+0.6 night)** **2024B/2025A**
“Identify X-ray Bright Quasars to Constrain the AGN Coronae”
- **BOK 2.5m BCSpec (2 night), Co-PI** **2024B**
“Redshifts of X-ray Bright Quasars to Constrain the AGN Corona”
- **MMT 6.5m Hectospec (0.3+0.3 night, 335 sources)** **2022B & 2023A**
“Complete the Hectospec Spectroscopic Survey of JWST NEP Time-Domain-Field”
- **MMT 6.5m Binospec (0.1+0.1+0.2 night)** **2022A & 2022B & 2023A**
Monitoring a Coronal Line Event AGN
- **MMT 6.5m Binospec (0.4 night, 6 sources)** **2023A**
Identify X-ray Bright Quasars and Constrain the AGN Coronal
- **SAO FLWO 1.5m FAST (0.2 night)** **2023A**

Measure the Black Hole Mass of an X-ray Bright Quasar to Constrain Its Coronal Properties

- **SAO FLWO 1.2m Keplercam** (1+1+2 night, g, r, i) **2023A & 2022B & 2022A**
“Monitoring the Continuous Optical Flares of a Coronal Line Event”
- **Sub-mm (3 tracks)**
- **Submillimeter Array (SMA)** standard science observation (3 tracks) **2022B**
“Monitoring with SMA a Highly Variable Flat Spectrum Radio Quasar in the JWST North Ecliptic Pole Time-Domain Field”

Selected Accepted Scientific Proposals as co-I

- **DECam 2025B (Survey, 56 nights):** PI: Weicheng Zang **2025**
- **NuSTAR Cycle 11+12 (Large, 880+800 ks):** PI: Laura Borrelli **2025/2026**
- **VLA 2026A (30 hours):** PI: Arran Gross **2025**
- **NuSTAR Cycle 10 (Large, 750 ks):** PI: Giorgio Lanzuisi **2024**
- **HST Cycle 22 (24 hours):** PI: Brent Smith **2024**
- **NuSTAR Cycle 8+9 (Large, 855+900 ks):** PI: Francesca Civano **2022/2023**
- **NuSTAR Cycle 5 (Large, 500 ks):** PI: Stefano Marchesi **2019**

Collaboration & Professional Service

- Member of **COSI** Collaboration **2026-**
- Member of **NuSTAR** Observation Quality Assurance Group **2025-**
- Member of **Broad-Band X-ray Mission** Science Analysis Groups **2025-**
- Member of **HEROIX** Mission Working Group **2025-**
- Member of **PRIMA** AGN Across Cosmic Time Working Group **2025-**
- Key Member of **HEX-P** Mission Black Hole Growth & Corona Working Group **2022-**
- Member of **AXIS** Mission AGN & Time-Domain Working Group **2022-**
- Member of **JWST PEARLS** Working Group **2022-**
- Member of **NuSTAR** Extragalactic Survey Team **2020-**
- Member of **Athena/NewAthena** Mission AGN Science Working Group **2020-**
- Organizers of CfA High Energy Astrophysics Division Seminar **2021-2023**
- + Reviewer for **NuSTAR, Chandra, Swift, CFHT** Observation Proposals
- ✦ Reviewer for ApJ, A&A

Selected Invited Talks

- Westlake University, Seminar **May 2026**
- Peking University, KIAA-DoA Seminar **May 2026**
- Tsinghua University, Colloquium **Dec 2025**
- Institute of High Energy Physics, Seminar **Dec 2025**
- MIT, High Energy Group Seminar **Oct 2025**

CfA, High Energy Seminar	Oct 2025
Columbia University, Theoretical High Energy Astrophysics Seminar	Oct 2025
CCA, Flatiron Institution, SPA Seminar	Oct 2025
Clemson University, Group Seminars	Apr 2025
Caltech, Tea Talk	May 2024
Zhejiang University, Colloquium	Sep 2023
Peking University, KIAA-DoA Seminar	Aug 2023
Tsinghua University, Departmental Seminar	Aug 2023
Yale University, Galaxy Lunch Talk	Apr 2023
MIT, Brown Bag Lunch Talk	Apr 2023
NASA GSFC, X-ray Astrophysics Laboratory AGN Seminar (Virtual)	Feb 2023
CfA, High Energy Seminar	Feb 2023
Arizona State University, Cosmology Seminar	Dec 2022
University of Arizona, Steward Observatory/NOIRLab Galaxy group seminar	Dec 2022
MIT, High Energy Astro Group Seminar (Virtual)	Apr 2022
INAF OAS, Bologna, X-ray group seminar	Sep 2019

Selected Contributed Talks

248th AAS Meeting (Press Conference)	Pasadena, Jun 2026
High Energy Astrophysics Division 22nd Meeting	St Louis, Oct 2025
AXIS Community Science Conference	Maryland, Apr 2025
High Energy Astrophysics Division 21st Meeting	Texas, Apr 2024
243rd AAS Meeting	New Orleans, Jan 2024
High Energy Astrophysics Division 20th Meeting	Hawaii, Mar 2023
241st AAS Meeting	Seattle, Jan 2023
<i>NuSTAR</i> 2022 Conference	Italy, June 2022
New England Regional Quasar and AGN Meeting	Connecticut, May 2022
High Energy Astrophysics Division 19th Meeting (<i>Poster</i>)	Pittsburgh, Mar 2022
238th AAS Meeting	Virtual, June 2021
237th AAS Meeting	Virtual, Jan 2021
235th AAS Meeting	Honolulu, Jan 2020
X-ray Astronomy 2019 Meeting (<i>Poster</i>)	Bologna, Italy, Sep 2019
High Energy Astrophysics Division 17th Meeting (<i>Poster</i>)	Monterey, Mar 2019
233rd AAS Meeting	Seattle, Jan 2019

Mentoring & Assistant Experience

Supervision of Caltech undergraduate student Z. Granadoz	2026
Supervision of University of Cambridge undergraduate student W. Chen	2026
Supervision of Imperial College London undergraduate student Y. Hu (Stanford Graduate)	2025

Co-supervision of Clemson graduate student R. Silver (NASA Postdoc Fellow)	2019-2025
Co-supervision of Clemson graduate student A. Pizzetti (ESO Fellow)	2019-2024
Co-supervision of Clemson undergraduate students D. Cole and Z. Hu	2019
Research Assistant, Clemson	2018-2020
Teaching Assistant (PHYS 2230), Clemson	2016-2017

Outreach

- The Silk Road Cameleers Series (Introduce AGN to Undergrads) Remote, **Apr, 2024**
- † Volunteer to teach astronomy and mathematics to elemental and high school students in the rural area of China Qiajia, **Summer, 2023**
- † Translate Sensing Dynamic Universe project into Chinese (help people with visual disability accessible to the dynamic Universe with sonified astromical light curves and spectra) **2022-2023**

References

- **Marco Ajello**, PhD supervisor, *Clemson University*, majello@g.clemson.edu
- **Francesca Civano**, Postdoc supervisor, *NASA Goddard*, francesca.m.civano@nasa.gov
- **Martin Elvis**, Postdoc co-supervisor, *CfA*, melvis@cfa.harvard.edu
- **Fiona Harrison**, Postdoc supervisor, *Caltech*, fiona@srl.caltech.edu
- **Stefano Marchesi**, PhD co-supervisor, *INAF-Bologna*, stefano.marchesi@inaf.it
- **Yue Shen**, Postdoc supervisor, *UIUC*, shenyue@illinois.edu
- **Daniel Stern**, Postdoc co-supervisor, *JPL*, daniel.k.stern@jpl.nasa.gov

Publication List

A total of **42** peer-reviewed papers, **6** submitted papers, **6** telescope white papers [ADS](#)

- **First-Author Papers**

- 10) **X. Zhao**, Elias Kammoun, Marco Ajello, et al., Submitted to AAS journals
Probing AGN Coronae with High-redshift AGN: PACHA
- 9) **X. Zhao**, Marco Ajello, Francesca Civano, et al., 2026, ApJ, 1000, 28
Transient Relativistic Iron Emission Line from a Flaring Supermassive Black Hole
- 8) **X. Zhao**, Luca Comisso, S. Marchesi, et al., 2025, ApJL, 995, L72
Fast X-ray Variability from the Coronae of Supermassive Black Holes
- 7) **X. Zhao**, S. Marchesi, M. Ajello, et al., 2024, ApJ, 975, 24
An X-ray Significantly Variable, Luminous, Type 2 Quasar at $z = 2.99$ with a Massive Host Galaxy
- 6) **X. Zhao**, F. Civano, C. N. A. Willmer, et al., 2024, ApJ, 965, 188
PEARLS: The NuSTAR and XMM-Newton extragalactic surveys of the JWST North Ecliptic pole Time-Domain Field II
- 5) **X. Zhao**, F. Civano, F. M. Fornasini, et al. 2021, MNRAS, 508, 5176
The NuSTAR extragalactic surveys of the JWST North Ecliptic pole Time-Domain Field
- 4) **X. Zhao**, S. Marchesi, M. Ajello, et al. 2021, A&A, 650, A57
The properties of the AGN torus as revealed from a set of unbiased NuSTAR observations
- 3) **X. Zhao**, S. Marchesi, M. Ajello, et al. 2020, ApJ, 894, 71
A broadband X-ray study of a sample of AGNs with [OIII] measured inclinations
- 2) **X. Zhao**, S. Marchesi, M. Ajello, 2019, ApJ, 871, 182
Compton-thick AGN in the NuSTAR Era. IV. A Deep NuSTAR and XMM-Newton View of the Candidate Compton-thick AGN in ESO 116-G018
- 1) **X. Zhao**, S. Marchesi, M. Ajello, et al. 2019, ApJ, 870, 60
Compton-thick AGNs in the NuSTAR Era. II. A Deep NuSTAR and XMM-Newton View of the Candidate Compton-thick AGN in NGC 1358

- **Significantly Contributed Papers and Mentored Students* Papers**

- 15) *R. Silver, F. Civano, **X. Zhao**, Submitted to ApJ
PEARLS: X-Ray Variability Analysis of the JWST North Ecliptic Pole Time-domain Field
- 14) *R. Silver, et al. (including **X. Zhao**), Accepted to ApJ
Compton-thick AGN in the NuSTAR Era. XI. Analyzing 11 CT-AGN candidates selected with ML
- 13) V. Paliya, S. Marchesi, **X. Zhao**, 2026, ApJ, 999, 83
A Serendipitous NuSTAR Detection of a Giant Radio Source Harboring an Obscured AGN
- 12) *R. Silver, F. Civano, **X. Zhao**, 2026, ApJ, 998, 200
PEARLS: NuSTAR and XMM-Newton Extragalactic Survey of the JWST North Ecliptic Pole Time-Domain Survey III
- 11) *Y. Hu, et al. (including **X. Zhao**), 2025, ApJ, 995, 189
Hard X-ray Contribution in AU Mic Flares and Its Minor Role in Atmospheric Escape

- 10) *A. Pizzetti, et al. (including **X. Zhao**), 2025, ApJ, 979, 170
Hydrogen column density variability in a sample of local Compton-thin AGN II
- 9) N. Torres-Albà, M. Stefano, **X. Zhao**, et al., 2023, A&A, 678, A154
Hydrogen Column Density Variability in a sample of local Compton-thin AGN
- 8) *R. Silver, N. Torres-Albà, **X. Zhao**, et al., 2023, A&A, 675, A65
A New Mid-Infrared and X-ray Machine Learning Algorithm to Discover Compton-thick AGN
- 7) *R. Silver, N. Torres-Albà, **X. Zhao**, et al. 2022, ApJ, 940, 148
Compton-thick AGN in NuSTAR Era. IX: joint NuSTAR and XMM-Newton analysis of four local AGN
- 6) *A. Pizzetti, et al. (including **X. Zhao**), 2022, ApJ, 936, 149
A multi-epoch X-ray study of the nearby Seyfert 2 galaxy NGC 7479: Linking column density variability to the torus geometry
- 5) S. Marchesi, **X. Zhao**, N. Torres-Albà, et al. 2022, ApJ, 935, 114
Compton-Thick AGN in the NuSTAR era VIII: A joint NuSTAR-XMM-Newton monitoring of the changing-look Compton-thick AGN NGC 1358
- 4) *R. Silver, N. Torres-Albà, **X. Zhao**, et al. 2022, ApJ, 932, 43
Chandra Follow-up Observations of Swift-BAT-selected AGNs II
- 3) N. Torres-Albà, S. Marchesi, **X. Zhao**, et al. 2021, ApJ, 922, 252
Compton-thick AGN in NuSTAR Era VI: The Observed Compton-thick Fraction in the Local Universe
- 2) S. Marchesi, M. Ajello, **X. Zhao**, et al. 2019, ApJ, 882, 162
Compton-thick AGNs in the NuSTAR Era. V. Joint NuSTAR and XMM-Newton Spectral Analysis of Three “Soft-gamma” Candidate CT-AGNs in the Swift/BAT 100-month Catalog
- 1) S. Marchesi, M. Ajello, **X. Zhao**, et al. 2019, ApJ, 872, 8
Compton-thick AGNs in the NuSTAR Era. III. A Systematic Study of the Torus Covering Factor

- Peer-reviewed Telescope White Papers

- 4) J. García, et al. (including **X. Zhao**), 2024, Front. Astron. Space Sci., 1471585
The High Energy X-ray Probe (HEX-P): Science Overview
- 3) F. Civano, **X. Zhao**, P. Boorman, et al., 2024, Front. Astron. Space Sci., 1340719
HEX-P: X-ray population contributing to peak of the Cosmic X-ray background
- 2) E. Kammoun, et al. (including **X. Zhao**), 2024, Front. Astron. Space Sci., 1308056
The High Energy X-ray Probe (HEX-P): Probing the physics of X-ray corona in active galactic nuclei
- 1) P. Boorman, et al. (including **X. Zhao**), 2024, Front. Astron. Space Sci., 1335459
The High Energy X-ray Probe (HEX-P): Probing the circum-nuclear environment in AGN down to extremely low luminosities

- Co-Author Papers

- 24) H. Yang, et al. (including **X. Zhao**), 2026, Submitted to AAS journals
A Minute-Cadence Deep Bulge Survey: First Data Release of DREAMS
- 23) R. Ortiz, et al. (including **X. Zhao**), Submitted to A&A
X-ray AGN in Their Host Galaxies: A Multiwavelength View of Concurrent Star Formation and Supermassive Black Hole Accretion within the NEP-TDF

- 22) Giorgio Lanzuisi, et al. (including **X. Zhao**), Submitted to A&A
The WISSHFUL program: the highest redshift UFO discovered in a non-lensed QSO
- 21) V. Elvezia Gianolli, et al. (including **X. Zhao**), Submitted to AAS journals
The hidden variability of the torus in local Active Galactic Nuclei: 20 years of Chandra, XMM, and NuSTAR observations
- 20) C. Karwin, et al. (including **X. Zhao**), Accepted to ApJ
Characterizing the Gamma-ray Emission from Low-Luminosity AGN
- 19) S. Willner, et al. (including **X. Zhao**), Submitted to AAS journals
PEARLS: JWST Counterparts of Micro-Jy Radio Sources in the NEP TDF. II. All Four Spokes
- 18) K. Imam, et al. (including **X. Zhao**), Accepted to ApJS
Source Identification for the Swift-BAT 150 Month Hard X-ray Catalog using Observations from Soft X-ray missions
- 17) H. Yang, et al. (including **X. Zhao**), 2026, AJ, 171, 151
KMT-2025-BLG-1616Lb: First Microlensing Bound Planet From DREAMS
- 16) I. Pal, et al. (including **X. Zhao**), 2026, ApJ, 998, 293
Unveiling Obscured Accretion in the Local Universe
- 15) I. Cox, et al. (including **X. Zhao**), 2026, ApJ, 997, 147
A systematic search for AGN obscuration variability in the Chandra archive
- 14) S. Creech, et al. (including **X. Zhao**), 2025, ApJ, 995, 203
The NH Distribution of Hard X-ray Selected Active Galactic Nuclei in the NEP Field
- 13) A. Banerjee, et al. (including **X. Zhao**), 2025, ApJL, 988, L50
Contemporaneous X-ray and Optical Polarization of EHSP Blazar H 1426+428
- 12) D. Sengupta, et al. (including **X. Zhao**), 2025, A&A, 697, A78
A Multi-Wavelength Characterization of the Obscuring Medium at the Center of NGC 6300
- 11) N. Torres-Albà, et al. (including **X. Zhao**), 2025, ApJ, 981, 91
Swift-XRT and NuSTAR Monitoring of Obscuration Variability in Mrk 477
- 10) I. Cox, et al. (including **X. Zhao**), 2025, ApJ, 979, 130
Chandra Follow-up Observations of Swift-BAT-Selected AGNs III
- 9) N. S. Khatiya, et al. (including **X. Zhao**), 2024, ApJ, 971, 84
Characterizing the γ -ray Emission from FR0 Radio Galaxies
- 8) R O'Brien, et al. (including **X. Zhao**), 2024, ApJS, 272, 19
TREASUREHUNT: Transients and Variability Discovered with HST in the JWST North Ecliptic Pole Time Domain Field
- 7) I. Cox, et al. (including **X. Zhao**), 2023, ApJ, 958, 155
A simple method to predict N_H variability in active galactic nuclei
- 6) S. P. Willner, et al. (including **X. Zhao**), 2023, ApJ, 958, 176
PEARLS: JWST counterparts of micro-Jy radio sources in the Time Domain field
- 5) C. N. A. Willmer, et al. (including **X. Zhao**), 2023, ApJS, 269, 21
PEARLS: Near Infrared Photometry in the JWST North Ecliptic Pole Time Domain Field
- 4) Q. Yang, et al. (including **X. Zhao**), 2023, ApJ, 953, 61

Probing the Origin of Changing-look Quasar Transitions with Chandra

- 3) D. Sengupta, et al. (including **X. Zhao**), 2023, A&A, 676, A103
Compton-thick AGN in the NuSTAR Era IX: Analysis of seven local CT-AGN candidates
- 2) R. A. Windhorst, et al. (including **X. Zhao**), 2023, AJ, 165, 13
Webb's PEARLS: Prime Extragalactic Areas for Reionization and Lensing Science: Project Overview and First Results
- 1) A. Traina, et al. (including **X. Zhao**), 2021, ApJ, 922, 159
Compton-Thick AGN in the NuSTAR era VII: a joint NuSTAR, Chandra and XMM-Newton analysis of two nearby, heavily obscured sources

- **First-author Telescope White Papers**

- 2) X. Zhao, Y. Shen, AXIS White Paper
AXIS-NEXUS: One Field for both SMBH Evolution and Time-Domain Study
- 1) X. Zhao, Y. Shen, PRIMA White Paper
PRIMA-NEXUS: One Field for both Galaxy Evolution and Time-Domain Study